Assignment 1

Course	Programming Fundamentals
Instructor	Ms. Hina Iqbal
Section	BCS-1E, BCS-1F

Important Guidelines:

1. If you use **AI tools** (ChatGPT, Copilot, etc.) or **copy code from other students/online sources**, your assignment will be given **zero marks** without exception.

   Prophet Muhammad (SAW) said: "Truthfulness leads to righteousness, and righteousness leads to Paradise" - *Sahih Al Bukhari 6094*

2. Submit your assignment **on time**. Late submissions will incur a 25%-mark deduction if submitted within 6 hours, a 50%-mark deduction if submitted within 24 hours, and submissions beyond 24 hours will result in a score of zero.
3. No marks will be awarded to individuals who do not submit the file on Google Classroom (GCR).
4. Only submit your cpp files on google classroom. **Do not zip your files.** Each cpp file should be renamed according to the format: **l25xxxx_q1.cpp. Failure to follow this format will result in a score of zero.**

5. In case of confusion please feel free to contact on email ***l227910@lhr.nu.edu.pk*** **or** ***hina.iqbal@nu.edu.pk***

Marks Distribution

Question	**Marks**
Q1. Meal Cost Calculator	20
Q2. Discount	20
Q3. Loyalty Points	15
Q4. Meal Deal Eligibility	20
Q5. Increment/Decrement Challenge	25
Q6. Cheapest Item Finder	15
Q7. Daily Sales Target	20
Q8. Combo Offer Checker	25
Total	**160**

Submission Instructions

1. Submit **only C++ source files (.cpp)** for each question.
2. File Naming Format:
    - Use your **roll number** followed by question number.

- Example:
    - `l25xxxx_q1.cpp`
    - `l25xxxx_q2.cpp` …

Mention your **Name and Roll Number** as a comment at the top of each file:

```
// Name: Ali Khan
// Roll No: L227910
// Q2 - Discount Program
```

Programming Fundamentals – Assignment 01

Maximum Marks: 160

Case Study: University Canteen Management Helper

The university canteen wants a **small utility software** that can help them manage **student meal orders, discounts, and billing**. You are required to implement the following subtasks using only:

- **Basic Input/Output** (`cin`, `cout`)
- **Variables & Data Types**
- **Operators (arithmetic, relational, increment/decrement, precedence)**
- **If–Else (including compound statements)**

Tasks

Q1. Meal Cost Calculator (20 marks)

The canteen sells **3 meal items**:

- Burger → 200 PKR
- Pizza Slice → 150 PKR
- Sandwich → 100 PKR

A student enters the **quantity of each item** they want.

- Calculate the **total cost**
- Display the final bill

Example Output:

```
Enter Burgers: 2
Enter Pizza Slices: 1
Enter Sandwiches: 3
Total = 850 PKR
```

Q2. Apply Discount (20 marks)

Discount policy:

- If **total bill > 500**, apply **10% discount**.
- If **total bill > 1000**, apply **20% discount**.
- Otherwise, no discount.

Extend Q1 to calculate and display the **discounted price**.
Use **relational operators** carefully to check conditions.

Example Output:

```
Enter Burgers: 2
Enter Pizza Slices: 1
Enter Sandwiches: 3
Total before discount = 850 PKR
```

```
Discount = 85 PKR
Final Bill = 765 PKR
```

Q3. Loyalty Points (15 marks)

The canteen gives **loyalty points**:

- For every **100 PKR** in the **final bill**, the student earns **1 point**.
- Use **integer division** and operators to calculate points.

Example Output:

```
Final Bill = 960 PKR
Loyalty Points Earned = 9
```

Q4. Student Eligibility for Meal Deal (20 marks)

The canteen runs a special deal:

- If a student buys **at least 2 Burgers AND 1 Pizza Slice**,
- OR if they spend **more than 700 PKR** (before discount),
  → They are eligible for a **free drink**.

Write a program to check **eligibility** using **compound if–else conditions** and display a message.

Example Output:

```
Burgers = 2, Pizza = 1, Sandwich = 0
Total = 550
Eligible for Free Drink
```

Q5. Increment/Decrement Challenge (25 marks)

The canteen manager gives a bonus scheme:

- Every time a student buys **more than 5 sandwiches**, increase their loyalty points by **2 (using increment operator)**.
- If the bill (before discount) is exactly **1000**, reduce loyalty points by **1 (using decrement operator)**.

Update your program from Q3 to handle this adjustment.

Example Output:

```
Total Bill = 1000 PKR
Loyalty Points = 10
After Adjustment = 9
```

Q6. Cheapest Item Finder (15 marks)

A student buys **3 items** (burger, pizza slice, sandwich) with given quantities.

- Calculate the cost of each type of item.
- Use **relational operators** to find and display which item **costs the least**.
- If two items have the same cost, display both.

Example Output:

```
Burgers = 1, Pizza = 2, Sandwich = 3
Burger Cost = 200
Pizza Cost = 300
Sandwich Cost = 300
Cheapest Item(s): Burger
```

Q7. Daily Sales Target (20 marks)

The canteen has a **daily sales target of 5000 PKR**.

- Input total sales made (sum of all bills for the day).
- If sales are **greater than or equal to target**, display `"Target Achieved"`.
- Otherwise, display how much more is needed:
  `"Target not achieved. Short by X PKR"`.

Example Output:

```
Enter today's sales: 4200
Target not achieved. Short by 800 PKR
```

Q8. Combo Offer Checker (25 marks)

The canteen offers a **combo meal deal**:

- If a student buys **1 Burger + 1 Pizza Slice + 1 Sandwich**,
  → Price is **400 PKR flat** (instead of 450).
- Otherwise, use the **regular calculation** from Q1.

Write a program to:

- Check if the student's order matches the **combo deal exactly**.
- Display bill accordingly.

Example Output:

```
Burgers = 1, Pizza = 1, Sandwich = 1
Combo Applied!
Final Bill = 400 PKR
```

Good Luck 🙂